DRILL 16

ROPES & RIGGING

NFPA 1001 Level 1 (JPR 4.3.20) / WorkSafeNB Reg 91-191

FIREGROUND KNOTS & TOOL HOISTING

⏱ **Time:** 45–60 min | 🚒 **Apparatus:** Any Engine/Truck | 🦺 **PPE:** Structural Gloves or Work Gloves, Helmet

👥 **Roles:** Drill Leader (Facilitator) | Safety Officer (Watcher) | Ground Rigging Crew & Elevated Haul Crew

🎯 OBJECTIVE & RURAL FIREGROUND VALUE

Every firefighter must be able to tie the 4 core fire service knots (**Bowline, Clove Hitch with safety, Figure-8 on a Bight, Half Hitch**) with gloved hands, and rig/hoist an axe, a Halligan bar, a ground ladder, and an attack hoseline up to a second-story window or roof without tools slipping or falling.

📦 TOOLS & APPARATUS SETUP

- 100-ft (30m) utility rope bag (minimum 1/2-inch nylon or polyester kernmantle rope).
- Tools to hoist: Flathead axe, Halligan bar, pike pole, 14-ft roof ladder, 50-ft section of 1.75" hose with nozzle.
- Elevated hoist station (station mezzanine, roof edge, hose tower, or 2nd story window).
- Webbing or edge roller/tarp to protect rope from sharp masonry parapets.

⚠ SAFETY WATCH-OUTS & STOP RULE

⚠ **Drop Zone Safety:** Establish a clear 5-metre (15-foot) **Drop Zone** directly below the hoist opening. No personnel may stand beneath hoisted tools.

⚠ **Tag Line Requirement:** A secondary tag line must be attached to the base of heavy tools or ladders to guide them clear of siding, windows, and obstacles during lifting.

⚠ **Rope Abrasion:** Always place a hose roller, canvas tarp, or turnout coat under the rope at the roof/window edge to prevent sheath damage.

🛑 **THE STOP RULE: Shout "STOP!" immediately if a knot begins to slip, if anyone enters the drop zone, or if the rope rubs bare over a sharp metal flashing.**

PHASE 1: THE CORE 4 KNOTS (15 MINUTES) — SLOW DRY RUN

1. **Bowline:** The premier rescue/loop knot. Forms a secure loop that will not slip or tighten under load.
2. **Clove Hitch:** Forms a tight grip around cylindrical objects (handles, pipes, hose). Always lock with an **Overhand Safety Knot** on the working end.
3. **Figure-8 on a Bight:** Quick, super-strong loop in the end or middle of a rope; easy to untie after heavy loading.
4. **Half Hitch:** Supporting/stabilizing knot used in combination with a clove hitch or bowline to keep long tools vertical while hoisting.

DRILL 16: PRACTICAL EVOLUTIONS & TAILGATE DEBRIEF

Ropes & Rigging • Practical Application

PHASE 2: HOISTING TOOLS STEP-BY-STEP (20 MINUTES) — BUDDY COACHING

1. **Hoisting an Axe (Axe Head DOWN):**
   - Tie a **Clove Hitch** (with safety) around the axe handle adjacent to the head.
   - Pull the standing rope along the handle toward the butt.
   - Cast a **Half Hitch** around the end of the handle.
   - Result: When pulled upward, the axe hangs strictly vertical with the heavy head pointing down.
2. **Hoisting a Pike Pole / NY Hook (Head UP):**
   - Tie a **Clove Hitch** (with safety) around the base of the handle.
   - Cast a **Half Hitch** around the center of the handle.
   - Cast a second **Half Hitch** around the hook/head of the tool.
   - Result: Hoisted point-up; hook cannot snag on gutters or window sills.
3. **Hoisting a Ground Ladder:**
   - Tie a **Figure-8 on a Bight** to create a 3-foot loop.
   - Pass the loop through the rungs 1/3 of the way down from the tip, wrap around beams, and loop over the top rungs.
   - Attach a secondary rope as a tag line to the bottom rung.
   - Hoist vertically with tip up.

PHASE 3: HOISTING A CHARGED VS UNCHARGED HOSELINE (15 MINUTES) — TIMED / STRESS

1. **Uncharged Line:** Fold 3–4 feet of hose back onto itself over the nozzle. Tie a **Clove Hitch** with safety around the fold and nozzle, and cast a **Half Hitch** around the hose 2 feet back.
2. **Charged Line:** Tie a **Clove Hitch** around the hose 1 foot behind the nozzle coupling. Pass the bight through the nozzle bail handle and loop over the tip. The rope pulls against the coupling, keeping the nozzle bail clamped closed during the lift.

⚠️ COMMON PITFALLS & FIREGROUND HAZARDS

- ❌ *Forgetting the overhand safety knot:* A clove hitch tied on a wooden or fiberglass tool handle can roll and untie itself under cyclic bouncing without a safety knot!
- ❌ *Hoisting axe with head pointing UP:* If hoisted head-up, the axe blade hooks on window sills, gutters, and siding, creating severe hang-ups.
- ❌ *Failing to use a tag line:* A dangling ladder or halligan swinging in the wind will shatter first-floor windows as it is hauled upward. Always use a ground firefighter on a tag line!

🗣️ 5-MINUTE TAILGATE DEBRIEF

1. ***"Did anyone struggle tying knots while wearing structural gloves?", "Was the drop zone strictly maintained?"***

✅ APPARATUS RESET CHECKLIST

- ☐ Utility ropes inspected along entire length for frays, flat spots, or chemical burns.
- ☐ Rope re-bagged into throw bag (stuffing method, never coiled).
- ☐ Hand tools inspected and remounted on apparatus.
- ☐ Attendance roster signed.

DRILL LEADER SIGNATURE	SAFETY OFFICER SIGNATURE	DATE COMPLETED	TRAINING HOURS